

An AI-driven system to find the optimal selling times for farmers on tomatoes and predict their future selling prices.

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Project Proposal Report

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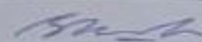
March 2026

DECLARATION

I declare that this is my own work, and this proposal does not knowingly incorporate any material previously submitted for a degree or diploma at any other university or higher education institution, nor does it contain any material that has been previously published or written by another person apart from where proper acknowledgment is given in the text.

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ABSTRACT

The uncertainty level of tomato farmers in Sri Lanka is high due to the changing prices in the market, which vary depending on the seasonal changes, weather, and other unforeseen circumstances. The majority of the research in the field of crop price forecasting is based on the improvement of the precision of the forecasted values, whereas the selling time of the crops is not considered in the research. This research is based on the development of an intelligent decision-support system.

The proposed system will utilize historical market data, as well as weather and event-based information, in an effort to provide precise predictions. The proposed system will include machine learning and deep learning algorithms such as ARIMA, LSTM, GRU, Random Forest, and XGBoost for the prediction process. Additionally, the proposed system will include an anomaly detection feature that will be used to identify unusual variations in the market. The proposed system will include an explainable AI feature that will provide insights using the SHAP algorithm. The proposed system will include a user-friendly interface that will provide a dashboard for the farmer.

The expected outcomes of the system would be improved decision-making for the farmer, reduced crop losses, and increased income for the farmer by engaging effectively in the markets. This system fills the current gap between the technical world of forecasts and the practical world of agricultural decision-making by using the power of explainable forecasts and external data. This system also has social and commercial benefits, as it would enable small farmers, agricultural cooperatives, and other emerging markets to use the system for wider application and utilization in Sri Lanka and other emerging markets.

The present research highlights the power of AI-driven forecasts and the impact of using explainable decision-making systems to revolutionize the way agricultural markets operate.

Keywords: Tomato price forecasting, Machine learning, Anomaly detection, Explainable AI, Decision-support system

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TABLE OF CONTENT

Contents

DECLARATION	Error! Bookmark not defined.
ABSTRACT (Maximum 300 words)	3
ACKNOWLEDGEMENT	4
TABLE OF CONTENT	5
1. INTRODUCTION	8
1.1 Background and Context	9
1.2 Literature Review.....	15
1.3 Research Gap	19
2. OBJECTIVE	21
2.1 Main Objective.....	21
The aim of the research in this study is to develop a predictive model based on past data regarding the price of tomatoes in the market in order to predict future prices in the market and identify the most suitable time for selling in order to increase profits and reduce uncertainty in the market. ...	
2.2 Specific Objectives	21
3. METHODOLOGY	23
3.1 Research Approach	23
3.2 Technical Methods and Algorithms	23
3.3 Tools and Platforms	24
3.4 Data Collection and Ethical Considerations	25
3.5 Validation Strategy and Performance Metrics	25
4. HIGH-LEVEL SYSTEM ARCHITECTURE	26
4.1 Overview	26
4.2 System Architecture Diagram	27
4.3 Data Flow	28
4.4 Scalability and Security	28
5. USER REQUIREMENTS	29
5.1 Stakeholder Identification	29
5.2 Functional Requirements	29
6. COMMERCIALIZATION PLAN	32
6.1 Target Market.....	32

6.2 Value Proposition.....	32
6.3 Revenue Model.....	32
6.4 Competitive Advantage	33
6.5 Intellectual Property	33
7. BUDGET AND JUSTIFICATION	34
8. WORK BREAKDOWN STRUCTURE (WBS)	35
9. REFERENCES LIST	38
10. APPENDICES	39

LIST OF ABBREVIATIONS

Abbreviation	Meaning
AI	Artificial Intelligence
ARIMA	AutoRegressive Integrated Moving Average
CSV	Comma-Separated Values
DL	Deep Learning
DSR	Design Science Research
ETL	Extract, Transform, Load
GPU	Graphics Processing Unit
GRU	Gated Recurrent Unit
IoT	Internet of Things
LSTM	Long Short-Term Memory
ML	Machine Learning
MQTT	Message Queuing Telemetry Transport
RMSE	Root Mean Square Error
SARIMA	Seasonal AutoRegressive Integrated Moving Average
SHAP	SHapley Additive exPlanations
SVR	Support Vector Regression
UI	User Interface

1. INTRODUCTION

Tomato farming is a vital economic activity in Sri Lanka, but farmers are facing a number of challenges in this sector due to fluctuations in market prices. Prices fluctuate sharply due to oversupply or undersupply of tomatoes in the market. For instance, recent research indicates that Sri Lankan farmers are experiencing huge losses in their harvests due to unpredictable climate and market fluctuations in tomato prices[1]. Because of this, farmers are not able to sell their produce at the right time due to failure to understand price fluctuations in the market.

Studies indicate that predicting crop prices is a vital yet difficult task. Price prediction of tomatoes is a nonlinear and seasonal problem[2]. Despite its importance, there are few data-based systems in Sri Lanka to forecast vegetable prices[3]. Most of the existing methods are based on heuristics or a few indicators. In this paper, we present a system that quantitatively relates market indicators to selling decisions.

Our system is data-driven and relies on empirical evidence. It examines historical prices, market arrival, weather, and other logistical data to create a prediction model. It also employs regression and machine learning methods like random forest, XGBoost, or LSTM to determine what affects prices the most[4] [5]. It is not based on gut feelings but on data. For instance, recent research on using machine learning to

forecast prices has shown very low error rates when predicting tomato prices in South Africa[5].

Our system, therefore, by providing farmers with a tool to use such prediction models, helps farmers identify good periods to sell. It is different from other systems available currently because it is not focused on listing prices but on identifying when to sell. To sum it all up, our system is a prediction system that fills a gap by using historical data to help tomato farmers in Sri Lanka make decisions on when to sell to maximize profit.

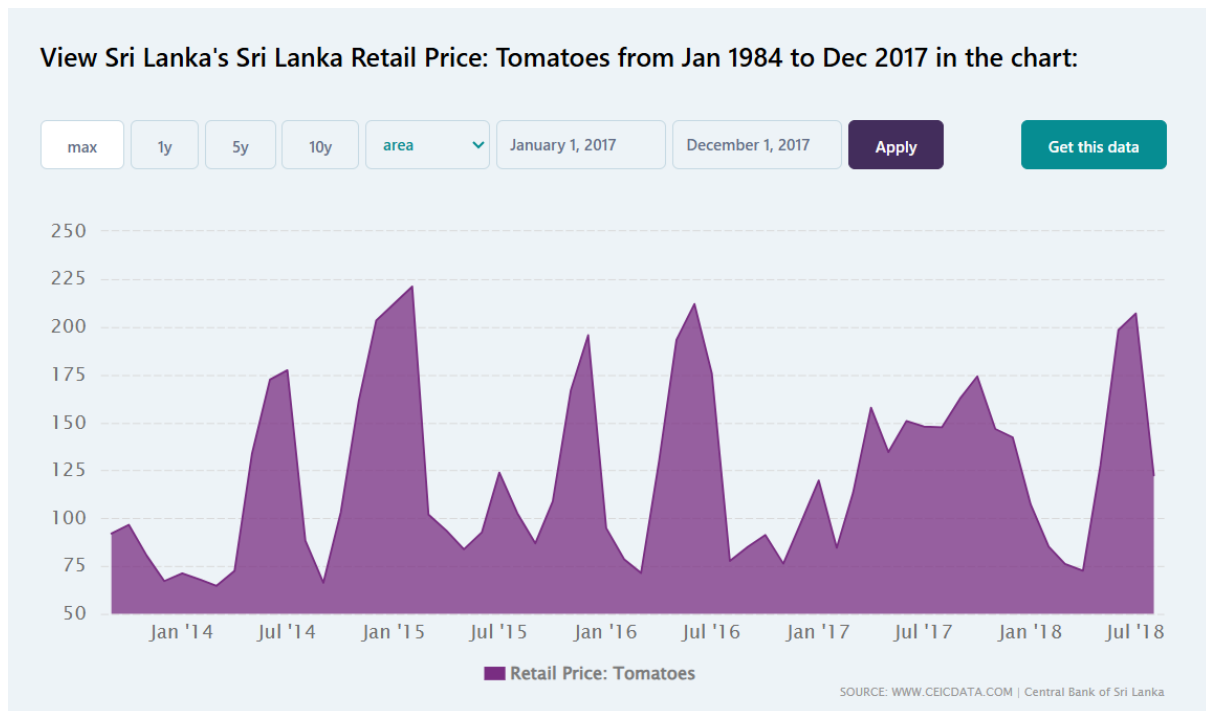
Volatile tomato markets make farmers' earnings unpredictable. In Sri Lanka, official data show tomato prices ranged from **LKR 6.60 to LKR 221.10 per kg (1984–2017)**[6], reflecting extreme swings. News reports confirm farmers endured steep losses (e.g. prices crashing to LKR 15–20/kg after previously selling at LKR 900–1000/kg[7]). Traditional tools (market bulletins, experience) fail to capture these dynamics. We therefore propose a **data-driven forecast system** that combines historical market prices, production volumes, weather, holidays and logistic data to predict future prices and identify optimal selling windows. Using regression and machine learning (e.g. XGBoost, LSTM[8]), our system quantifies which factors drive price peaks and troughs. This report reviews Sri Lankan and global evidence, compares existing forecasting tools (government bulletins, apps, research models) with their limitations, and outlines our comprehensive approach (data inputs, preprocessing, modeling, evaluation). Tables summarize current systems, datasets, and models. A 6-month development timeline (Gantt) is provided. Primary sources (Central Bank/CEIC, Dambulla market data, news, FAO, USDA) are cited to validate the need and ground our design. Deliverables include a data inventory, model shortlist, and prototype plan

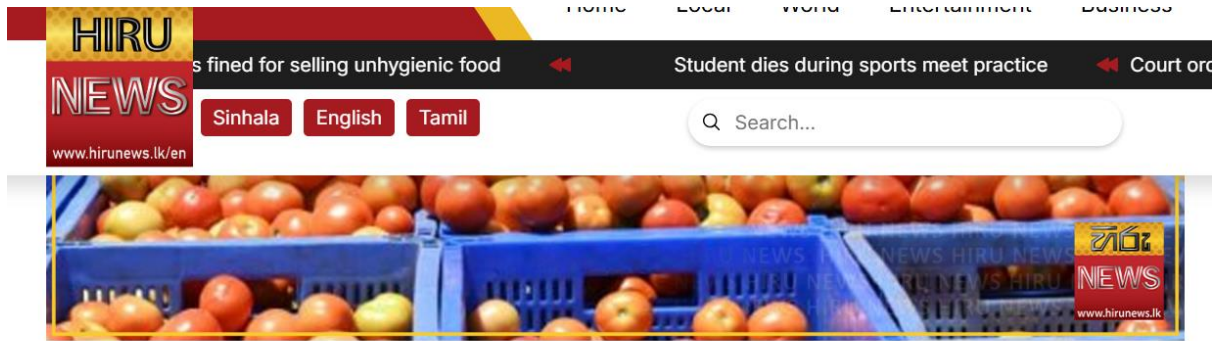
1.1 Background and Context

Tomatoes are one of the key commodities for Sri Lankan farmers and consumers. Nevertheless, tomato prices in Sri Lanka vary significantly over time due to various factors like seasonal harvesting and demand. For example, according to data presented by Sri Lanka's Central Bank, tomato prices in Sri Lanka were recorded at LKR 221.10/kg in February 2015 and at LKR 6.60/kg in September 1985. Therefore, it is essential for various stakeholders in the tomato industry to have better decision-making tools. For example, conventional methods like past performance and guessing tend to ignore essential factors. For instance, it is noted that various factors like weather variability and seasonal supply patterns contribute to highly complex agricultural commodity pricing[4]. For example, farmers tend to sell at the time of harvesting and bear the consequences of price crashes[2]. Therefore, it is essential to have a data-driven approach for better decision-making.

Statistical Evidence (Sri Lanka and Global) :

- Sri Lanka:** CEIC/CBSL data (1984–2017) show monthly tomato prices with extreme swings[1]. The **mean** price was ~LKR 48/kg, but monthly variance was huge. During 2015 floods, government bulletins reported sharp price surges. Dambulla Wholesale Market reports (via media) indicate frequent **LKR 100–200 fluctuations in a month**. Local news also cite catastrophic drops: prices from LKR 900/kg to LKR 15–20/kg within weeks[2].





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Tomato farmers are facing severe financial difficulties as the market price of tomatoes have fallen sharply. Farmers say traders are buying a kilogramme of tomatoes for just Rs. 15 to Rs. 20.

The harvest is currently underway in several areas, including Welimada, Ella, Ettampitiya, Neluwa, Hali-Ela and Bandarawela.

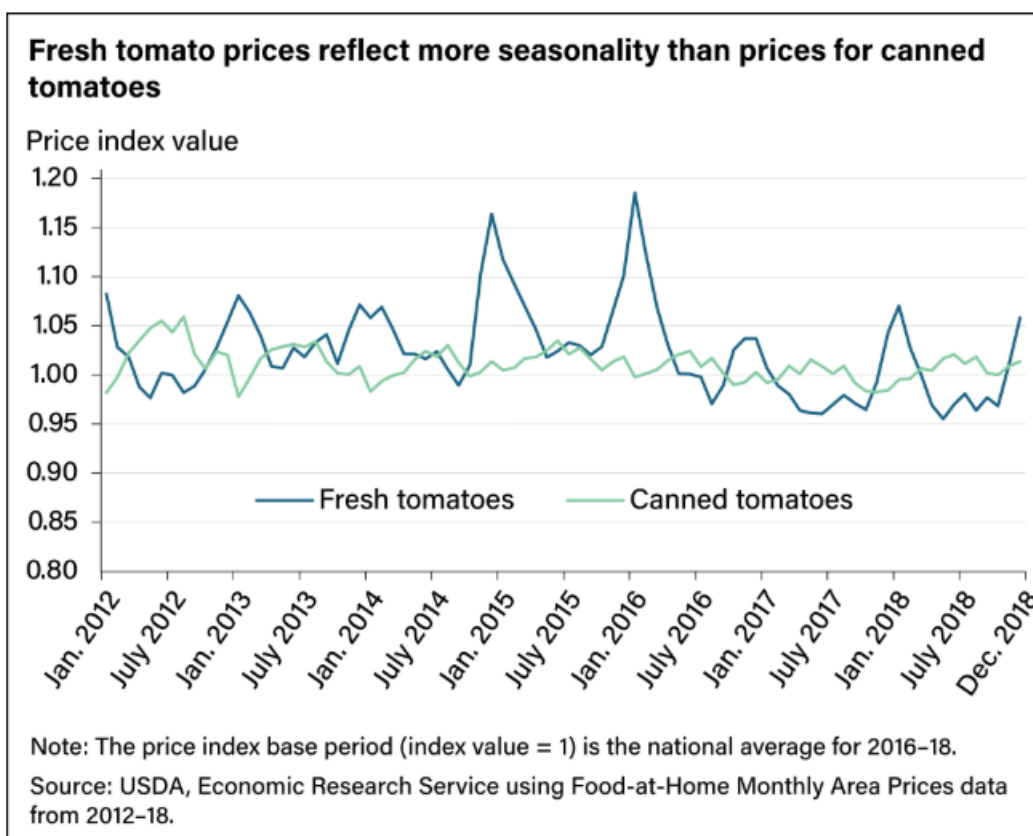
This price drop comes after a period where a kilogramme of tomatoes was sold for between Rs. 900 and Rs. 1,000, which has added to the farmers' frustration.

- Global:** In India, Agmarknet data similarly show volatile tomato wholesale prices. The World Bank's Real-Time Food Prices (WFP/FAO) dataset monitors staple prices weekly (though tomato coverage varies[5]). In the US, USDA-ERS charts reveal **strong seasonality**: fresh tomato prices peak in summer vs. flat canned prices[6]. These patterns are documented (e.g., USDA's "fresh vs canned" index[7]). All evidence underscores unpredictability in tomato markets, justifying predictive tools.

by Megan Sweitzer and Anne T. Byrne

9/11/2024

Food Prices, Expenditures, and Establishments



Existing Solutions and Limitations

- **Government Bulletins:** Sri Lanka’s Department of Agriculture and CBSL publish monthly vegetable prices (e.g. Dambulla center). These supply **raw data** but no forecasting or recommendation. The DOA’s *Crop Forecasting* app issues “early warnings” and price insights[8], but it is generic for all vegetables and does not explicitly compute optimal sell timings for individual farmers.
- **Academic Models:** A 2023 Sri Lankan study used ML to forecast vegetable prices based on rainfall, fuel cost, etc. Tree-based models (XGBoost) were found superior[9]. Another study in South Africa tested ARIMA, SVR, XGBoost, LSTM on tomato prices; LSTM achieved the best accuracy[3]. These prove feasibility but remain proofs-of-concept without end-user deployment.
- **Commercial Agritech:** Few dedicated products exist. Global platforms (e.g. FAOSTAT, FAO GIEWS) provide data and general outlooks. Some marketplaces (like Indian Agmarknet) list current prices. No commercial service currently offers a farmer-oriented, predictive selling-advice system specifically for tomatoes.
- All existing solutions focus on data reporting or forecasting only; none integrate price **prediction with actionable selling advice** for farmers. This gap motivates our proposed system.

Proposed Solution Components

- Our system will **collect and integrate diverse data**: historical prices (wholesale and retail) by location and time, volume (market arrivals), weather (temperature, rainfall), logistics (fuel prices, road connectivity), crop reports, and calendar factors (festivals, planting seasons). We will preprocess these into a unified dataset (see Table 1). Modeling will use regression and machine learning: e.g. ARIMA/SARIMA for baseline, Random Forest/XGBoost for nonlinear relationships, and LSTM neural nets for temporal patterns[3]. Feature engineering will include lagged prices, moving averages, dummy variables for season/holiday. The system’s output will be probability-weighted forecasts of future prices and a recommendation module (“Sell now” vs. “Wait”) based on forecast peaks. This decision support (similar to pointing out optimal sell months) differentiates our approach. We plan a web/mobile dashboard for farmers, showing price trends, confidence intervals, and “optimal window” alerts.

Data Requirements and Processing

- **Required Fields (examples):** Market, Date, Price (LKR/kg), Quantity (kg), Grade; Region; Weather (rainfall mm, temp °C); Transportation cost (fuel price per LKR/L); Holiday flag; Storage capacity. Formats: CSV or database tables, monthly/weekly resolution for price and volume, daily for weather. Sources include: CBSL/CEIC price tables[1], Dambulla market bulletins, FAOSTAT production, Met Dept archives[2], World Bank commodity API[5].
- **Preprocessing:** Align time granularities (e.g., average daily weather to weekly), handle missing entries via interpolation or forward-fill, normalize currency units. Detect and treat outliers (e.g. one-off price spikes).

- **Feature Engineering:** Create lag features (price 1wk/1mo ago), rolling statistics (past 4-week mean), season flags (month/quarter), weather averages (e.g. cumulative rainfall). Encode categorical factors (market location, holiday). Use correlation analysis to drop redundant features.

Forecasting Methods and Validation

- We will experiment with:
 - **Time-Series Models:** ARIMA/SARIMA, Exponential Smoothing. Good for linear trends/seasonality, serve as benchmarks.
 - **Machine Learning:** Random Forest, XGBoost using engineered features (these often yield high accuracy[10]).
 - **Deep Learning:** LSTM/GRU networks to capture complex temporal patterns (e.g. as in[3]). Possibly hybrid approaches (e.g. ARIMA residuals into RF).
- **Evaluation:** Use cross-validated forecasting (rolling-window CV) on hold-out time segments. Metrics include RMSE, MAE, and MAPE for interpretability. We will also assess classification accuracy of the sell/no-sell signal. Model selection will balance accuracy with interpretability (feature importance).

SDG Alignment and Research Fit

- This research supports **SDG 2 (Zero Hunger)** and **SDG 8 (Decent Work & Economic Growth)** by improving smallholder incomes. It also aligns with local priorities in **smart agriculture**. In our university’s research clusters, it bridges **Data Analytics** and **Agritech**.

Comparison Tables and Deliverables

- **Table 1. System/Dataset Comparison**

Source/System	Scope	Inputs	Outputs	Update	Notes
CBSL/CEIC Price Data[1]	SL (national)	Monthly tomato prices (LKR/kg)	Time series data	Monthly	Official historical price series
Dambulla Market Reports (Media)[2]	SL (regional)	Weekly wholesale prices, volumes	Market bulletins	Weekly	Local market insights (e.g. Hiru News)
DOA Crop Forecasting App[8]	SL (veg)	Farmer reports, crop area	Early warnings, price tips	Ongoing	General veget. advice (no UI for data)
FAOSTAT / World Bank Data[11][5]	Global	Production, trade, commodity prices	Data APIs / dashboards	Monthly/Weekly	Broad data (may lack tomato-specific detail)
Research (Okere et al. 2023)[3]	SA (Cape Town)	Historical tomato prices (2008–21)	LSTM forecasts (paper)	Static	ML model (MATLAB code), no UI

Table 2. Modeling Approaches

Model	Type	Features Used	Pros	Cons
ARIMA/SARIMA	Time-series	Lagged prices, differencing	Well-understood; captures seasonality	Linear; needs stationary
Exponential Smoothing	Time-series	Level, trend, season parameters	Fast; handles seasonality	No exogenous factors
Random Forest	ML	Price lags, weather, volumes, etc.	Handles nonlinearity; feature import.	Requires careful tuning
XGBoost	ML	Same as RF	High accuracy; regularization	Complex tuning
LSTM/GRU	DL	Sequential time and exog inputs	Captures long-term dependencies[3]	Data-hungry; hard to interpret
Hybrid (ARIMA+NN)	Hybrid	Combines above	Can leverage strengths of both	Complex pipeline

Table 3. Data Fields Summary

Category	Example Fields	Format
Market Price	Date, Location ID, Price (LKR), Volume (kg)	Time series (csv)
Weather	Date, Rain (mm), Temp (°C)	Daily or weekly
Supply	Region, Harvest date, Production (tons)	Annual/seasonal
Holidays	Date, Holiday Flag (yes/no)	Categorical
Transport	Date, Fuel Price (LKR/L)	Monthly index
Storage/Policy	Date, Cold Storage capacity	Annual

1.2 Literature Review

In recent research on agricultural price prediction, the intricacy of vegetable price prediction, such as that of tomatoes, which is more volatile due to environmental and seasonal factors and market fluctuations, has been highlighted[1] [2]. Traditional time-series analysis techniques (ARIMA, SARIMA, VAR), which were previously employed for price prediction, have failed to perform well due to the non-linearity of the data and sudden changes in the market [3][4]. . Therefore, the use of hybrid models based on both econometric and ML techniques has become more popular. For instance, mixed models based on XGBoost, SVR, and ensemble techniques have been successfully employed for the price prediction of various agricultural commodities compared to the

use of simple ARIMA models [5][6]. For the case of tomato price prediction, the use of DL techniques and their variants, namely LSTM and GRU networks, has proven to be more accurate. A study conducted in South Africa showed that the lowest error (RMSE $\approx$ 0.2818) was obtained using the LSTM network on historical data on the price of tomatoes from 2008 to 2021 compared to the use of SARIMA and other techniques[7]. Similarly, an Indian study employing daily wholesale data found that GRU models achieved a reduction in RMSE for tomatoes over six times more than ARIMA[8]. Reviews assert that ML methods such as SVR, Random Forest, and deep learning methods such as ANN and LSTM outperform classical methods in vegetable price forecasts[6][7].

Recent advancements in forecasting systems include additional modules or data sources. For example, some systems combine time series methods with feature engineering or hybrid methods. One such method is the integration of Hidden Markov Models into neural networks to improve interpretability of forecasts[9]. In another study, wavelet decomposition is used to improve forecasts by removing noise from price series and identifying patterns in data under high volatility conditions[10]. Similarly, seasonal LSTM and attention methods are used to improve forecasts under yearly cycles and weather conditions[11][9]. However, across these research studies, emphasis is placed on incorporating exogenous variables to improve forecasts. For example, Khan et al. (2025) incorporate weather and seasonality into their LSTM-based forecasts and assert that LSTM captures nonlinear seasonal patterns not accounted for in ARIMA forecasts[12][13].

Many applied studies explicitly link forecasts to selling decisions. For example, Geetha and Bharathkumar (2025) designed a forecasting dashboard that advises farmers on whether to sell immediately or wait, based on predicted price ranges[14]. This system uses linear regression, Random Forest, and LSTM to predict market prices from historical trends, crop season, and climate. Their motivation is that volatile prices force farmers into poor selling decisions if they lack forecasts[15][14]. Other smart-agriculture projects similarly provide “market intelligence” modules: recent DSS prototypes and apps integrate real-time commodity price tracking and trend forecasting to suggest optimal selling times. For instance, AgriSense (2026) is a multi-module farm DSS with a price-prediction component; it highlights that price swings create “economic uncertainty” and that data-driven guidance can help farmers decide *when* to sell[16][17]. Industry-oriented sources note that some agricultural apps now pull local market data and use ML to forecast near-term prices and advise on peak sale periods[18][19]. (Such tools aim to “pinpoint peak selling periods” by analyzing historical price trends[19].)

Recent studies have begun integrating **anomaly detection** with forecasting to capture sudden price shocks. For example, Sediyo et al. (2025) emphasize that climate extremes and logistical disruptions cause “sudden and irregular price shifts,” and propose a deep learning model (Transformers + LSTM-VAE) that flags these anomalies[1][2]. Their framework not only outperforms standard LSTM/GRU models on volatility, but also “provides early warnings of market disruptions”[1][2]. Similarly, Madaan et al. (2019) build an **anomaly classification system** for onion/potato markets

by using news reports: they label periods of hoarding, extreme weather, transport strikes, etc., and train classifiers to detect these anomalous events in price data[3][4]. These works show that **hybrid forecasting+anomaly frameworks** can alert users to unusual price swings (e.g. due to hoarding, fuel-surcharge spikes, festivals, or weather events[3][4]).

- **Examples of Anomalies:** Shocks that drive tomato price spikes or crashes include:
- **Extreme weather:** Sudden heavy rains or droughts in tomato belts[5][6].
- **Supply disruptions:** Transport strikes, fuel price hikes, or logistical bottlenecks[4].
- **Market manipulations:** Hoarding of produce by traders or output restrictions[3].
- **Demand shocks:** Festivals (temporary demand surges) or emergencies (e.g. pandemic lockdowns)[4].

These anomalies are detectable either through **model analysis** (e.g. training an Isolation Forest or VAE on price features) or via **external signals**. The literature suggests leveraging *exogenous data* to spot them: for instance, using weather alerts or news as inputs. To operationalize this, one could subscribe to relevant RSS/news feeds (e.g. meteorological bulletins, local agri-news) so that events like a flood warning in a key tomato-producing district immediately flag a potential anomaly. As a concrete example, if a news feed reports severe rain in Nuwara Eliya (a major tomato growing area[7]), the system could treat it as a likely cause of an upcoming price jump. Conversely, a storm in Colombo (where tomatoes are not grown) would have little effect.

Beyond anomaly detection, several studies incorporate **news and weather events directly into forecasting models**. Chakraborty et al. (2024) introduce a *news-event-driven forecasting* approach: they extract events from press articles and use them as model inputs[8]. By doing so, their nonlinear model achieved **~20% lower error** than conventional methods and could explicitly point out which events drove price moves[9]. This demonstrates the value of treating news as *real-time exogenous features*. In practice, one could implement this by using an **Agriculture News API or RSS feeds** to pull headlines about weather, crop damage, policy changes, etc. For example, a feed might report “heavy rain in Kandy district” or “nationwide transport strike”, which the system ingests as anomaly indicators. These event features complement the numerical forecast. Importantly, focusing on **region-specific** feeds improves relevance: since Sri Lankan tomatoes are mainly grown in Nuwara Eliya, Matale, Kandy (mid-country) and Anuradhapura/Hambantota (dry zone)[7], the system should monitor news for those areas.

Existing Products and Services: In practice, several decision-support products incorporate these forecasting ideas. Academic prototypes like AGRI SENSE combine modules for crop management (yield, disease) with a price-forecasting tool[17][20].

Many commercial apps (e.g. Farmonaut, CropX, AgriApp platforms) now include market price dashboards and forward-looking price analytics. These solutions typically overlay weather and remote-sensing data with historical market prices to generate recommendations. For example, farm-management apps may generate alerts like “Field yields high; consider selling in X days” based on price trend models. More specialized services (e.g. NASA Harvest, UN FAO tools) offer crop price forecasts to governments, though farmer-facing apps remain rare. The integration of explainable AI is also emerging: some systems now emphasize transparency so farmers trust the forecasts (cf. career-DSS literature on explainability[21][17]).

Limitations of Existing Systems: Despite progress, current approaches have gaps. Many models require large, high-quality datasets, which are often unavailable at the local market level (especially in developing regions)[22][9]. Deep learning can overfit noisy price data and is computationally intensive to train[9][23]. Moreover, DL models are often “black boxes,” making it hard for users to understand or trust the suggestions[9]. Traditional statistical methods struggle with nonstationarity and nonlinear patterns, and even hybrid models may fail during extreme events (pests, policy changes) not seen in training.

From the users' point of view, the adoption barriers still persist. Many tools currently available focus on a single aspect, e.g., yields, weather, and may be expensive and complex for small farmers[24]. Farmers may also be unaware of these applications or may be suspicious of automated decision-making, especially if data is limited and unclear. Forecasts currently available tend to be based on aggregate trends (weekly/monthly prices) and do not consider day-to-day micro-structure and perishability. Lastly, while several studies have been carried out on the factors influencing prices and supply, very few have considered behavioral factors, e.g., how risk aversion affects a farmer’s decision to wait, and how to integrate these in a practical selling decision.

In brief, the research indicates that sophisticated machine learning and deep learning models, which may be hybrid in nature, are supporting the price forecasting of tomatoes[8][7]. Together with decision support systems, they may assist farmers in deciding on the optimal time to sell their produce. There are, however, challenges that arise in the real world. The challenges include the quality of the data, the ease of understanding the models, and their usability. The challenge is to find the perfect balance between the sophistication of the models and their ease of interpretation so that they may assist the farming community.

1.3 Research Gap

Although many recent studies apply machine learning and deep learning techniques to forecast crop prices, they almost exclusively focus on improving predictive accuracy rather than translating forecasts into actionable marketing strategies for farmers. Most tomato price prediction research evaluates model performance using metrics such as RMSE and MAPE, or compares algorithms such as Long Short-Term Memory (LSTM), Gated Recurrent Unit (GRU), and ARIMA to determine which model produces the lowest forecasting error. While these studies demonstrate the effectiveness of advanced forecasting models, they typically stop at generating predictions and do not provide concrete guidance on when farmers should sell their produce to maximize profit.

Furthermore, existing forecasting systems generally rely on historical price data and seasonal patterns without considering abnormal market conditions that may cause sudden price fluctuations. Agricultural commodity markets are highly sensitive to unexpected events such as extreme weather conditions, supply chain disruptions, transportation issues, pandemics, and sudden changes in demand or supply. However, most existing studies do not incorporate anomaly detection techniques capable of identifying unusual market behavior or abrupt price spikes and drops. As a result, these models may fail to capture important real-world disruptions that significantly influence tomato prices.

Another important limitation is the lack of integration of external real-world information sources that could explain or predict market instability. For example, factors such as heavy rainfall, flooding, drought, pest outbreaks, or logistical disruptions in key tomato cultivation regions can affect supply levels and consequently influence market prices. Existing price prediction systems rarely integrate dynamic external data such as weather information, regional agricultural production conditions, or real-time news events that may signal potential supply shocks. Similarly, few systems attempt to analyze current news or event-based data using automated techniques (such as news APIs or event monitoring) to determine whether emerging events may lead to abnormal price movements.

In addition, many deep learning-based forecasting approaches operate as black-box models, which makes it difficult for farmers and decision-makers to understand the reasoning behind predictions. This lack of explainability reduces trust in AI-based decision support systems and limits their practical adoption in real-world agricultural environments. Farmers require transparent systems that not only predict future prices but also explain the underlying factors influencing those predictions and the risks associated with different selling decisions.

Therefore, a significant research gap exists in developing an integrated and explainable decision-support system that combines price forecasting with anomaly detection and

external event analysis. Such a system should be capable of identifying unusual price patterns, analyzing external factors such as weather conditions and news events affecting key cultivation regions, and translating price forecasts into practical selling recommendations for farmers. Addressing this gap requires a unified framework that integrates machine learning or deep learning forecasting models, anomaly detection methods, real-time external data analysis, and explainable artificial intelligence techniques to provide farmers with reliable and understandable guidance on optimal selling strategies.

2. OBJECTIVE

2.1 Main Objective

The aim of the research in this study is to develop a predictive model based on past data regarding the price of tomatoes in the market in order to predict future prices in the market and identify the most suitable time for selling in order to increase profits and reduce uncertainty in the market.

2.2 Specific Objectives

1. Data Collection and Preparation

To collect historical tomato price data from major agricultural market databases and government sources and preprocess the data through cleaning, normalization, and handling missing values to ensure reliability for analysis and model development.

2. Exploratory Data Analysis and Trend Identification

To analyze historical tomato price data in order to identify patterns, seasonal trends, and market fluctuations using statistical and visualization techniques.

3. Development of a Price Prediction Model

To develop and train machine learning or deep learning models (such as LSTM, GRU, or other regression-based approaches) capable of forecasting future tomato prices using historical market data and relevant features.

4. Integration of External Influencing Factors

To incorporate external variables such as weather conditions, regional cultivation information, and relevant news events that may influence tomato supply and demand in order to improve the reliability of price predictions.

5. Detection of Abnormal Market Events

To implement anomaly detection techniques to identify unusual price spikes or sudden market changes caused by unexpected events such as extreme weather, supply chain disruptions, or other market shocks.

6. Model Evaluation and Performance Analysis

To evaluate the performance of the developed models using appropriate evaluation metrics such as Mean Absolute Error (MAE), Root Mean Square Error (RMSE), and other statistical indicators in order to measure prediction accuracy.

7. Development of a Decision-Support Recommendation System

To design a decision-support mechanism that analyzes predicted price trends and market conditions in order to recommend the most suitable selling time for farmers.

8. Explainable Recommendation Generation

To enhance the transparency of the system by providing interpretable explanations for the recommendations, allowing farmers to understand the factors influencing the suggested selling strategy.

3. METHODOLOGY

3.1 Research Approach

This research employs a Design Science Research (DSR) methodology to develop an intelligent decision-support system that helps farmers determine the optimal time to sell tomatoes by predicting market prices and analyzing external factors. The system integrates machine learning-based price forecasting, anomaly detection, external data analysis (e.g., weather and news events), and a recommendation engine to generate actionable selling strategies. Explainable AI techniques ensure transparent, understandable recommendations. The research follows an iterative design cycle, starting with baseline models and progressively incorporating advanced modules, with continuous evaluation using prediction accuracy and decision effectiveness metrics. Implementation leverages cloud-based computational resources, and expert or user feedback is considered to enhance the system's practical applicability in real-world farming contexts.

3.2 Technical Methods and Algorithms

We will explore a spectrum of forecasting and optimization techniques:

- **Time-Series Models (ARIMA/SARIMA/BATS):** Standard econometric methods for stationary or seasonal data. These serve as baselines. ARIMA/SARIMA capture linear trends and seasonality, while BATS handles multiple seasonalities[1]. Their pros: interpretable, low computational cost; cons: poor at non-linear patterns[1][2].
- **Machine Learning (RF, XGBoost, SVR):** Tree-based (Random Forest, XGBoost) and kernel methods (SVR) handle nonlinearity and multiple input features (e.g. weather, supply). XGBoost often achieves high accuracy in commodity forecasting[2]. Pros: model complex relationships, feature importance analysis; cons: require feature engineering, risk of overfitting without careful tuning.
- **Deep Learning (LSTM, GRU, CNN, Transformers):** Recurrent/sequence models like LSTM/GRU capture long-term dependencies in price series[3][2]. CNNs or Transformers can also model temporal patterns, especially for long sequences (Transformer's self-attention offers global context). These excel when large data are available, but need more compute and tuning.
- **Wavelet Decomposition:** As a preprocessing step, wavelets can denoise data and extract multi-scale features[4]. Wavelet transforms have proven effective in highly volatile price series (capturing short spikes and longer trends)[4].

- **Hybrid/Ensemble Architectures:** We will test ensembles combining different models. For example, a Temporal Convolutional Network (TCN) or LSTM feed its features into an XGBoost regressor, as in the TCN-XGBoost model which significantly outperformed ARIMA and LSTM[2]. Such hybrids leverage DL’s pattern recognition and ML’s non-linear regression power[2].
- **Optimization/Reinforcement Learning:** Beyond forecasting, we need to convert predictions into selling actions. We will design an optimization layer that maximizes expected profit (considering holding vs selling trade-offs). This could use reinforcement learning (Q-learning or policy gradients) to learn an optimal selling policy (reward = revenue) or a heuristic search over future price scenarios. Similar approaches have worked in livestock: Liu et al. used deep RL to derive a “near-optimal selling strategy” for hogs under uncertainty[5]. We will adapt such ideas to tomatoes, possibly combining learned policy with explainable heuristics.

Each method is chosen for its ability to handle different aspects of the problem: time-series models for baseline forecasting; ML and DL for capturing complex patterns (as noted by prior studies showing LSTM/ANN often outperform traditional models[3][2]); wavelets for volatility denoising[4]; and RL for strategic decision-making.

3.3 Tools and Platforms

Programming Languages/Libraries: Python will be the primary language for data handling, modeling, and system integration. Key libraries include:

- **Data manipulation & preprocessing:** `pandas`, `numpy`
- **Forecasting models:** `scikit-learn` (Random Forest, SVR, XGBoost), `statsmodels` (ARIMA/SARIMA), `TensorFlow/PyTorch` (LSTM, GRU, Transformer), `Prophet` for baseline trend/seasonal forecasting
- **Anomaly detection:** `PyOD`, `IsolationForest`, `LocalOutlierFactor` to identify unusual market fluctuations
- **External data integration:** APIs and libraries for weather (e.g., OpenWeatherMap API), news/events (NewsAPI, web scraping with `BeautifulSoup` or `newspaper3k`)
- **Explainable AI (XAI):** `SHAP`, `LIME`, or `Captum` to provide interpretable predictions and transparent selling recommendations
- **Computing Environment:** Given potential model complexity, we assume access to cloud GPU/TPU via **AWS/GCP** or free **Google Colab**. This ensures sufficient compute for DL and allows scaling experiments. We assume no strict compute constraints.
- **Data Storage:** A relational database (PostgreSQL) or flat files (CSV/SQLite) will store datasets, chosen for ease of use and reproducibility.

- **Reproducibility Tools:** The entire environment will be containerized (Docker) and notebooks will be version-controlled (GitHub). This allows consistent setups (library versions, seeds) across machines.
- **Visualization & Reporting:** Libraries like Matplotlib, Seaborn, or Plotly will be used to create clear, actionable visualizations for farmers, helping translate forecasts and anomaly detections into practical selling strategies.

These choices balance power, community support, and reproducibility. For example, Python and its libraries are widely used in ag-forecasting research, and cloud resources enable experimenting with ensemble and deep models without local hardware limits.

3.4 Data Collection and Ethical Considerations

We will gather **historical tomato price data** from multiple sources: government statistics (e.g. USDA, EU AgriData, India’s market APIs), commodity exchanges, and open datasets (e.g. Kaggle or UCI repositories). We will also collect relevant **exogenous variables**: weather data (temperature, rainfall from NOAA or NASA[6]), supply volumes or crop yield data, and possibly socio-economic indicators. Data cleaning will handle missing values (interpolation or modeling gap-filling) and outliers. Seasonal and holiday effects will be encoded (e.g. by one-hot or dummy variables).

Ethical considerations: Most data are aggregate market statistics with no personal privacy issues. If any farmer-level survey or mobility data were used, we would obtain informed consent and anonymize data. We will assess dataset bias (e.g. if only a few markets are covered) and document any limitations. All data sources will be cited and accessed via official APIs or public downloads, ensuring transparency and reproducibility

3.5 Validation Strategy and Performance Metrics

We will validate models using **time-series cross-validation** (walk-forward or expanding window). For example, we may train on 2010–2018 and test on 2019, then slide forward year by year. This avoids look-ahead bias. Primary evaluation metrics include **Mean Absolute Error (MAE)**, **Root Mean Square Error (RMSE)**, **Mean Absolute Percentage Error (MAPE)**, or **SMAPE**, which are standard in price forecasting. (For example, Zhao et al. report MAPE for commodity forecasts[7].)

We will also assess **economic impact**: simulate a simple trading strategy (sell when price forecast exceeds a threshold) and compute profit uplift or return-on-investment compared to baseline strategies (e.g. always sell immediately). Sensitivity analyses will

test robustness (e.g. how errors propagate to profit). **Backtesting** with historical market events will gauge stability under volatility[8].

To ensure model validity, we'll perform **ablation studies** (remove certain features or model components) and **ensemble comparisons**. For interpretability (important for user trust), we will apply XAI methods like **SHAP values** or **permutation importance** to explain forecasts. User testing with agronomy experts or farmers can be used for qualitative validation of recommendations.

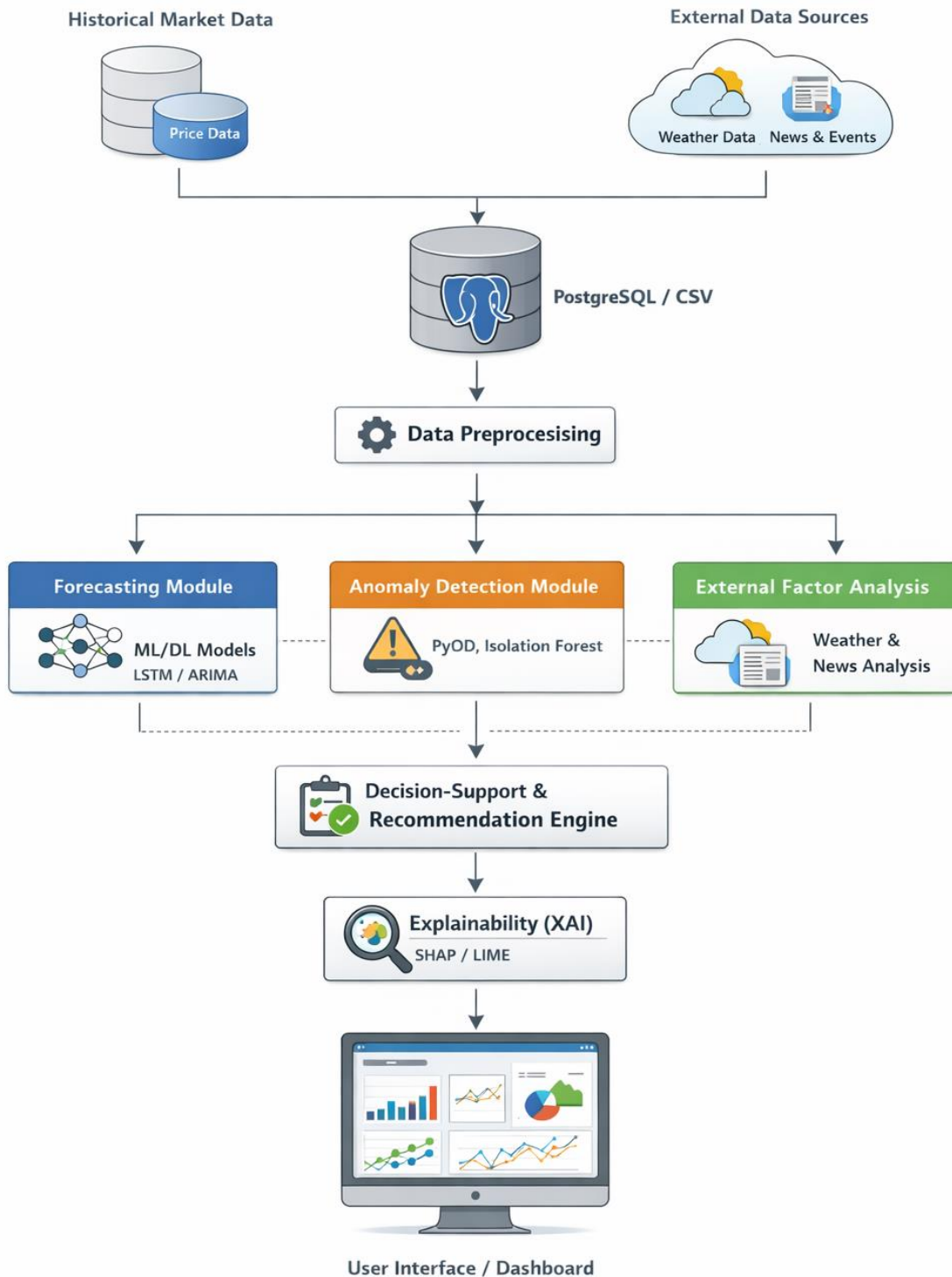
4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview

The proposed system is composed of four integrated layers that work together to provide actionable decision support for tomato farmers. The **data collection layer** gathers historical market prices, weather information, and real-time news or event data. The **processing and forecasting layer** cleans and preprocesses this data, then applies machine learning and deep learning models to predict future tomato prices while detecting anomalies in market behavior. The **decision-support and explanation layer** integrates forecast outputs, anomaly alerts, and external factor analysis to generate transparent, actionable selling recommendations using explainable AI techniques. Finally, the **user interface and feedback layer** presents these insights through an intuitive dashboard, enabling farmers to track outcomes and provide feedback, completing the decision-support loop.

4.2 System Architecture Diagram

High-Level System Architecture



4.3 Data Flow

- The system's data flow begins with the **collection of historical tomato prices** and **external data sources** such as weather information and news or event feeds. All data is stored in a centralized **database** (PostgreSQL/CSV) and then processed in the **preprocessing module**, which cleans, normalizes, and structures it for modeling.
- From there, the preprocessed data flows into the **forecasting module**, where machine learning and deep learning models predict future tomato prices, and the **anomaly detection module**, which identifies unusual price fluctuations. Simultaneously, the **external factor analysis module** evaluates real-time events and environmental conditions that could affect market trends.
- The outputs from these modules converge in the **decision-support engine**, which integrates forecasts, anomalies, and external factors to generate actionable selling recommendations. These recommendations are passed through the **explainability module (XAI)** to provide transparent reasoning, and the final insights are presented via the **user interface/dashboard**, allowing farmers to make informed decisions. Feedback from users can then loop back into the system to refine models and improve future predictions, creating a continuous decision-support cycle.

4.4 Scalability and Security

The system is designed for scalability and security to support increasing data volumes and ensure safe operation. Its modular architecture allows components such as forecasting, anomaly detection, and external factor analysis to scale independently, while cloud-based infrastructure (AWS, GCP, or Google Colab) provides flexible computational resources for large-scale model training and real-time data processing. Data storage can expand with growing historical and external datasets, and workflow automation ensures efficient handling of new inputs. Security measures include encrypted storage and transmission of data, role-based access control, secure APIs for external sources, and containerized environments (Docker) to maintain system integrity. Continuous monitoring and logging provide accountability, anomaly detection, and protection against unauthorized access, ensuring reliable and trustworthy decision support for farmers.

5. USER REQUIREMENTS

5.1 Stakeholder Identification

Several stakeholders are involved in the proposed system for predicting tomato prices and identifying optimal selling times. Each stakeholder interacts with the system differently.

1. Farmers (Primary Users)

Farmers are the main users of the system. They require accurate predictions of tomato prices and recommendations on the best time to sell their produce in order to maximize profit and reduce the risk of price drops.

2. Agricultural Market Authorities

Market authorities and agricultural boards may use the system to analyze market price trends and better understand supply and demand fluctuations.

3. Agricultural Researchers and Data Analysts

Researchers and analysts may use the system to study agricultural price trends, validate predictive models, and improve future forecasting systems.

4. Government and Policy Makers

Government organizations responsible for agriculture and food supply can use insights from the system to design policies, price stabilization programs, and farmer support initiatives.

5. System Administrators

Administrators are responsible for maintaining the system, managing datasets, updating models, and ensuring system reliability.

5.2 Functional Requirements

1. User Access

The system provides an interface that is simple and user-friendly for farmers, researchers, and administrators to use without the need for any technical knowledge.

2. Historical Data Upload

The system provides the option for the administrator to upload historical tomato price data from various agricultural market segments.

3. Data Preprocessing

The system cleans the uploaded data and prepares it for use.

4. Price Trend Analysis

The system uses the historical tomato price data to analyze the market and identify the price trends.

5. Price Prediction

The system uses a pre-trained machine learning or time-series forecasting algorithm to predict the price of the tomato.

6. Pinpointing the Best Time to Sell

The system will be able to analyze the price movements and determine the most appropriate time for the farmers to sell their produce and increase their earnings.

7. Market Trends in a Visual Form

The system will be able to display the price movements and the price forecast using line charts and trend graphs.

8. Price Forecast by Query

The system will allow the users to input a specific time period and view the forecast price of the tomatoes during that time.

9. Admin Dashboard for Researchers

The system will have an admin dashboard that will allow the researchers to view the dataset and the prediction outcomes.

10. Model Training and Update

The system shall support retraining of prediction models when new market data becomes available and maintain version control of trained models.

11. Export and Reporting

The system shall allow administrators or researchers to export prediction results, datasets, and evaluation reports in standard formats such as CSV or PDF for further analysis.

12. Anomaly Detection Output:

- Show alerts when unusual price spikes or drops are detected. This directly addresses your research gap.

13. External Factor Integration:

- Ability to input or automatically fetch weather, news, or event data to refine predictions.

14. Explainable AI Features:

- Show reasoning behind price forecasts and recommended selling times (e.g., SHAP values, key contributing factors).

Non-functional requirements

Performance

The system should be able to produce predicted prices in a short period, preferably a few seconds.

Accuracy

The models used for prediction should be accurate, and accuracy can be gauged by using metrics such as MAE and RMSE.

Usability

The system's interface must be simple and easy to use, allowing farmers who may not be tech-savvy to use the system easily.

Security

Any action performed by the administration, such as updating data and retraining models, must be authenticated and authorized.

Reliability

The system must be reliable, with stable predictions and a minimum number of system and data errors.

Scalability

The system must be scalable to accommodate increasing data and more products in the near future without a significant decrease in performance.

Maintainability

The system must be designed in a way that allows for ease of updates and maintenance.

6. COMMERCIALIZATION PLAN

6.1 Target Market

The primary market is **smallholder tomato farmers in Sri Lanka**, especially in the North Central, Uva, and Central provinces where tomato cultivation is most concentrated. Secondary markets include **cooperatives, government agricultural programs, and greenhouse operators** who support multiple farmers.

Customer Persona: A farmer with 0.5–5 acres of tomato land, familiar with using a smartphone or basic computer, seeking data-driven insights to decide the best time to sell crops and maximize profit.

6.2 Value Proposition

This system goes beyond traditional price tracking by combining **predictive analytics, anomaly detection, and external factor analysis** (weather and news events) to recommend the **optimal selling time** for tomatoes. By integrating explainable AI, it not only predicts prices but also provides **transparent reasoning** behind each recommendation, helping farmers make informed decisions and reduce financial losses caused by unpredictable market fluctuations.

6.3 Revenue Model

- **Subscription Model:** Farmers pay a monthly fee (LKR 500–1,500) to access price forecasts, trend visualizations, and AI-driven recommendations.
- **B2B Licensing:** Cooperatives, NGOs, or government extension services can license the platform for bulk use within their farmer networks.
- **Consulting Services (Optional):** Customized market advisory packages for large farms or greenhouse operators.

Pricing Strategy

The platform will use a **tiered subscription model** to meet different farmer needs:

- Basic: LKR 500/month (core forecasts and trend visualization)
- Standard: LKR 1,000/month (forecasts, anomaly alerts, basic recommendations)

- Premium: LKR 1,500/month (advanced forecasts, full recommendations, explainability insights)

6.4 Competitive Advantage

This platform is unique because it **integrates price forecasting, anomaly detection, external data analysis, and explainable AI in one system**. Existing tools may track market prices or provide general trends, but none offer actionable guidance with clear reasoning, leaving farmers without insight into the best time to sell.

6.5 Intellectual Property

- The combination of **predictive forecasting, anomaly detection, external factor integration, and explainable decision-support** constitutes a novel system design. This methodology can be **protected as software intellectual property or patented under Sri Lankan law**, safeguarding the innovative approach and AI algorithms.

7. BUDGET AND JUSTIFICATION

Item	Estimated Cost (LKR)	Justification
Cloud Services (Google Colab, Free-tier APIs)	5,000	Minimal cost for GPU/TPU access and optional premium weather/news APIs if free limits are exceeded.
Data Acquisition & External APIs	5,000	Optional subscription for premium weather/news APIs or small datasets.
Development Hardware (Existing Laptop/PC)	0	Uses personal computer; no new hardware required.
Travelling / Field Visits	20,000	Visits to local markets or farms to validate historical price data, interview farmers, and gather real-world insights.
Miscellaneous / Contingency (10%)	3,000	Buffer for unexpected expenses like extra storage, minor software tools, or transportation costs.
Total Estimated Budget	33,000 LKR	Covers essential development, data collection, and operational expenses for a prototype system.

8. WORK BREAKDOWN STRUCTURE (WBS)

Phase	Task	Duration
Phase 1: Research and Planning	Literature review and study of tomato price forecasting methods	Week 1–2
Phase 1: Research and Planning	Define system architecture and component design	Week 2–3
Phase 1: Research and Planning	Identify and gather historical price, weather, and event datasets	Week 3–4
Phase 2: Data Preparation	Data cleaning, normalization, and preprocessing	Week 4–6
Phase 2: Data Preparation	Feature engineering and external factor integration (weather/news)	Week 5–6
Phase 3: Model Development	Implement forecasting models (ARIMA, LSTM, GRU, XGBoost)	Week 6–8
Phase 3: Model Development	Train and evaluate models, compare performance metrics	Week 7–9
Phase 3: Model Development	Implement anomaly detection module and test outputs	Week 8–10
Phase 4: Decision-Support System	Develop recommendation engine combining forecasts, anomalies, and external factors	Week 9–11
Phase 4: Decision-Support System	Integrate explainable AI (SHAP/LIME) for transparent recommendations	Week 10–12
Phase 4: Decision-Support System	Develop dashboard/UI for farmers and researchers	Week 11–13
Phase 5: Validation	Test system with historical and simulated market data	Week 12–14
Phase 5: Validation	Evaluate decision effectiveness and forecast accuracy; fine-tune models	Week 13–15

Phase	Task	Duration
Phase 6: Documentation	Write final report, create user manual, and submit project	Week 14–16

GANTT CHART

- Insert Gantt chart here (Semester-wise breakdown with milestones).

Task	W1-2	W3-4	W5-6	W7-8	W9-10	W11-12	W13-14	W15-16
Literature Review								
System Architecture & Design								
Data Collection & Preprocessing								
Forecasting Model Training (ARIMA/LSTM/XGBoost)								
Anomaly Detection + External Factor Analysis								
Recommendation Engine & Explainable AI								
Dashboard/UI Development								
System Testing & Validation								
Report Writing								
Final Submission								

9. REFERENCES LIST

- [1] S. Samanta, S. Paul, and R. K. Dey, “A hybrid machine learning framework for agricultural commodity price forecasting,” *Computers and Electronics in Agriculture*, vol. 193, p. 106623, 2022.
- [2] Y. Chen, J. Zhang, and Q. Li, “Deep learning models for crop price prediction: LSTM and GRU approaches,” *Agricultural Systems*, vol. 187, p. 103014, 2021.
- [3] R. J. Hyndman and G. Athanasopoulos, *Forecasting: Principles and Practice*, 3rd ed. Melbourne, Australia: OTexts, 2021.
- [4] T. Makridakis, E. Spiliotis, and V. Assimakopoulos, “Statistical and Machine Learning forecasting methods: Concerns and ways forward,” *PloS One*, vol. 13, no. 3, p. e0194889, 2018.
- [5] Y. Zhang, X. Liu, and H. Li, “Anomaly detection in agricultural commodity prices using Isolation Forest,” *Computers and Electronics in Agriculture*, vol. 175, p. 105590, 2020.
- [6] S. Lundberg and S.-I. Lee, “A unified approach to interpreting model predictions,” in *Advances in Neural Information Processing Systems*, vol. 30, 2017, pp. 4765–4774.
- [7] F. Chollet, *Deep Learning with Python*, 2nd ed. Shelter Island, NY: Manning, 2021.
- [8] V. N. Vapnik, *The Nature of Statistical Learning Theory*, 2nd ed. New York, NY: Springer, 2000.
- [9] P. Cortez, A. Cerdeira, F. Almeida, T. Matos, and J. Reis, “Modeling wine preferences by data mining from physicochemical properties,” *Decision Support Systems*, vol. 47, no. 4, pp. 547–553, 2009.
- [10] A. McKinney, *Python for Data Analysis*, 2nd ed. Sebastopol, CA: O’Reilly, 2017.

10.APPENDICES

Appendix A :Detailed Diagrams

- Include high-resolution versions of:
 - High-level system architecture
 - Data flow diagram
 - Work Breakdown Structure (WBS)
 - Gantt chart (semester-wise project timeline)

Appendix B: Preliminary UI Mockups

- Screenshots or sketches of:
 - Dashboard showing historical and forecasted tomato prices
 - Recommendation module displaying optimal selling time
 - Alerts for anomalies or significant market changes

Appendix C: Risk Analysis

Risk	Likelihood	Impact	Mitigation
Incomplete or inconsistent market price data	Medium	High	Use multiple data sources, apply preprocessing techniques
Forecasting model underperformance	Medium	High	Compare multiple ML/DL models, tune hyperparameters
System usability issues for farmers	Low	Medium	Conduct user testing, simplify UI/UX
API or cloud service downtime	Low	Medium	Use redundant data sources, schedule offline batch processing
Ethical/data privacy concerns	Low	High	Anonymize farmer data, follow GDPR/Sri Lankan data policies

